

Almanac 47 for the HP Prime

Python version: the ALMANAC page and the horizon chart — user manual

Two graphic views of the C47 suite, drawn with the HP Prime's own graphics module (hprime): the “vintage” ALMANAC page and the horizon chart. The numbers come from nav.py and navdata.py — the same series and methods as the C47 programs (the Sun from the full VSOP87 series, valid 2000–2050, nutation, the 58 navigation stars, sunrise, twilight, Moon phase). On a PC every value on the screens was compared with the text page of nav.py and with the C47 ALMANAC screen: identical.

The Sun and the stars only: nav.py has no Moon position and no planets (the Moon phase and age are shown). The table has the Sun and the brightest stars higher than 10°, as on the C47.

It supports, and does not replace, the Nautical Almanac. Always cross-check the values.

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1. Files

File	From	What it is
nav.py, navdata.py	python/	the almanac computation and data (also a text menu)
main.py	python/hprime/	what the Python app runs: starts on the ALMANAC page
hplib.py	python/hprime/	text, symbols, keys, main loop
almview.py, skyview.py	python/hprime/	the two views

2. Copy to the calculator

- On the calculator: Apps, select **Python**, Save it as a new app, e.g. **Almanac47** (the original Python app stays as it is).
- With the **HP Connectivity Kit** (calculator on USB, or the Virtual Calculator): open the new app under Application Library and drag the six .py files into it (replace its main.py). If your Kit does not accept dropped files, create each file in the app's Symb view with the same name and paste the text.
- Start the app (Apps → Almanac47). From its shell, **import skyview** starts on the horizon chart.

3. Inputs

The program asks in the terminal, like the nav.py menu:

Input	Format	Example
Year, Month, Day	numbers	2026, 9, 26
UT	hh:mm or hh:mm:ss	14:57
Lat, Lon	“25 20” = 25° 20′, “25.5” = decimal degrees, “-” for S / W	25 20, 55 12

- Time is UT (UT1):** for a sight timed in UTC add DUT1 (under 0.9 s) — as in the almanac.
- Signs:** latitude N +, S –; longitude E +, W –.

4. Keys

Key	Action
UP / DOWN	one hour later / earlier
ENTER	switch ALMANAC ↔ HORIZON
LEFT / RIGHT	HORIZON: the previous / next body for the top line (shown in red)
ESC	exit (ON also stops a Python program)

5. The views

The previews were made on a PC with stand-ins for the hprime module (the same scripts; the calculator's own font may make text a little wider or narrower).

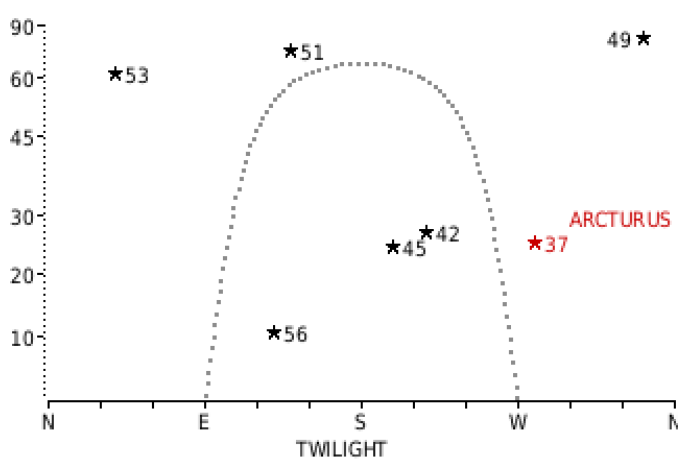
26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0

BODY	GHA	DEC	HC	ZN
ARIES	229 40.8			
☉ SUN	46 25.7 S 1 24.5	-11 06.4	273.7	
★37 ARCTURUS	15 27.7 N 19 02.7	24 59.7	280.2	
★49 VEGA	310 13.1 N 38 48.8	75 46.1	342.6	
★51 ALTAIR	291 39.3 N 8 56.5	69 23.0	140.4	
★42 ANTARES	341 55.1 S 26 29.5	27 00.7	217.3	
★56 FOMALHAUT	244 53.4 S 29 28.7	10 35.9	130.0	
★53 DENEb	279 05.4 N 45 22.8	61 13.8	039.3	
★45 SHAULA	325 49.4 S 37 07.5	24 29.0	198.3	

NAUT TWI 01:20 15:01 MOON 100% ● FULL
RISE/SET 02:09 14:12 AGE 15.5 DAYS
MER PASS 08:11 SD 16.0

UP/DOWN 1 HOUR ENTER HORIZON ESC EXIT
DOES NOT REPLACE THE NAUTICAL ALMANAC

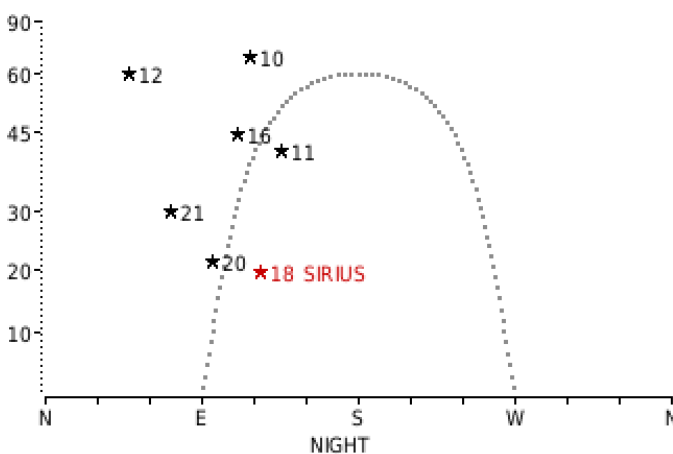
26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0



ALMANAC: date, UT, DR; GHA Aries; BODY, GHA, DEC, HC, ZN of the Sun and the stars; naut. twilight, rise/set, meridian passage, Sun SD, Moon phase and age. A negative Hc is white on black. 26 Sep 2026 14:57 UT, 25° 20' N 055° 12' E.

HORIZON: Hc up (sine scale), Zn across, the celestial equator dotted, the Sun and the stars with their numbers; the top line shows ZN and HC of the selected body. Same time and place.

23-09-2026 03:00 UT DR N 30 00.0 E 0 00.0



HORIZON at night: 23 Sep 2026 03:00 UT, 30° N 0° E (the Sun below the horizon).

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0

BODY	GHA	DEC	HC	ZN
ARIES	229 40.8			
☉ SUN	46 25.7 S 1 24.5	-11 06.4	273.7	
★37 ARCTURUS	15 27.7 N 19 02.7	24 59.7	280.2	
★49 VEGA	310 13.1 N 38 48.8	75 46.1	342.6	
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NAUT TWI 01:20 15:01 MOON 100% ● FULL
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UP/DOWN 1 HOUR ENTER HORIZON ESC EXIT
DOES NOT REPLACE THE NAUTICAL ALMANAC

6. The text menu

nav.py alone (Execute script, or **menu()** in the shell) is the text version: 1 Sun, 2 Star, 3 Rise/Set, 4 Moon phase, 5 Almanac (the same page as the ALMANAC view, as text).

7. Memory and size

- The six files are about 26 KB; the Prime's Python memory is far larger than needed.
- Text goes through the PPL command TEXTOUT_P (font size and colour), keys through GETKEY.

8. Limitations

- Series only, no almanac tables.** The Sun comes from the truncated VSOP87 series: valid 2000–2050, better than 0.1'. The tables TBL_1 / TBL_5 of the C47 and Free42 are not read (they mainly serve the Moon and the planets, which this version does not compute).
- No Moon position and no planets:** only the Moon phase and age. For the Moon and the planets use the C47, Free42 / DM42n (NAVFULL) or the Python version on the PC (python/native).
- ΔT fixed** at 69.2 s (TT – UT1), as in the C47 series; the input time is UT1 (add DUT1 to UTC).

- **Two views only:** ALMANAC and HORIZON (no split sky, all-sky, animation or star pages).
- **Memory** is not a limit on the Prime: the Moon and the planets could be added later.

9. Status

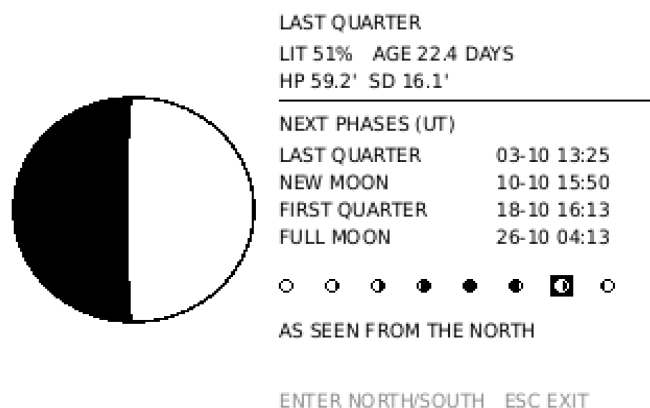
- Not yet tried on a real HP Prime. Checked on a PC (stand-in module) and in a MicroPython interpreter.
- Not sure on the real Prime: the exact pixel size of its fonts (columns may be a few pixels off) and whether ESC reaches GETKEY in every firmware (ON always stops the program).

10. MOON47: the Moon phase

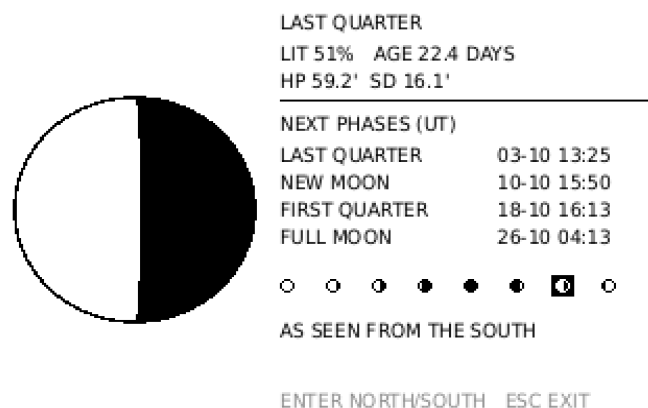
MOON47 is a separate program, the Moon's phase page without NAV, INIT or the tables:

python/hpprime/moon47.py (about 9 KB, one file). Make a new Python app (e.g. Moon47) and put this file in it as its main.py. It asks nothing: it takes the date and time from the Prime's clock.

03-10-2026 16:00 LT TZ=+4



03-10-2026 16:00 LT TZ=+4



3 Oct 2026 12:00 UT (16:00 LT, TZ=+4), from the north after Enter: from the south

- **The page:** the Moon as a disc (the lit part filled), the phase name, % lit, age in days, HP and SD, the next new Moon, first quarter, full Moon and last quarter (date and time UT), and the eight phase symbols with today's one inverted. The phase does not depend on where you are; the latitude only turns the picture.
- **Keys:** Enter turns the picture to the view from the south and back; Esc ends.
- **The clock is local time.** If it is not on UT, set the line TZ = 0 at the top of main.py to your offset in hours (4 for UT+4). With TZ = 0 the clock is taken as UT: the phase times stay right, the age and % lit are then a few hours off. The top line shows the clock: the time UT and TZ=0, or the local time LT and TZ (TZ=+4).
- **Accuracy** against the full series of NAV, 2000–2050: phase times and age within 4 minutes (most within 1), HP 0.03', SD 0.01', % lit 0.03 %: the 20 largest terms of the Moon (Meeus, ch. 47), the same formulas as MOON47 on the C47 (python/moon47.py; tests/test_moon47.py).